Geometric Integration with Event Detection for Simulating Coordinate-Free State-Space Rigid Body Dynamics with Impacts

This repo is an extension of another repo I created, `geometric_rigid_body_dynamics`, in which I demonstrated how a geometric integrator is used to simulate rigid body dynamics directly in terms of homogeneous transformation matrices, twists, and wrenches, without having to deal with multiple coordinate charts.

In this repo, I add two more features:

- event detection: detects when the rigid body impacts the ground;
- impact map: updates the state of the rigid body after impact.

I came across a preprint of a similar paper: "Lie Group Variational Collision Integrators for a Class of Hybrid Systems" by Tran and Leok. It takes a slightly different approach to the problem but is a helpful reference!

1 Hybrid Dynamics

We model and simulate a bouncing ball as a hybrid system, i.e., a system consisting of continuous dynamics (the differential equations of motion of the ball when in free fall) and discrete dynamics (the algebraic equations that relate the post-impact state of the ball to its pre-impact state). At a high level of abstraction, the hybrid dynamics are

$$\dot{x} = f(x), \qquad \text{if } h(x) \neq 0, \tag{1a}$$

$$x^+ = \Delta(x^-), \qquad \text{if } h(x) = 0, \tag{1b}$$

where f is a vector field that defines the continuous-time dynamics, $h(x)$ is the *event function* (a function of the state x that returns zero when impact occurs) and Δ is a function that maps the pre-impact state x^- to the post-impact state x^+.

In this repo, the event function h will return the height above ground of the lowest point on the ball, and the impact map Δ will arise from conservation of momentum. My focus in this repo is on how to incorporate event detection in a Runge-Kutta geometric integrator, rather than on the details of the event function, so I will start with a simple case: a spherical ball; as such, the distance between the ground and the bottom of the ball is not affected by the ball's orientation. Simulation of an ellipsoidal ball is likely to produce more interesting (maybe even chaotic!) results but I'll save that for later.

1.1 Continuous Dynamics

In this repo, I am using the geometric integrator proposed by Crouch and Grossman (J. Nonlinear Sci., 1993) to simulate the motion of a rigid body in 3D space. The body has 6 degrees of freedom (DOF), but in order to avoid the pitfalls associated with choosing a set of 6 generalized coordinates, I am using a coordinate-free approach by working directly with the 4x4 homogeneous transformation matrix that represents the body's configuration in space.

Denote this transformation matrix $T_{\mathrm{wb}} \in \mathbb{SE}(3)$, where $\{\mathrm{w}\}$ is the world frame and $\{\mathrm{b}\}$ is the body frame, i.e., a frame oriented along the body's principal axes of inertia and whose origin is located at the body's center of mass (COM). The special Euclidean group $\mathbb{SE}(3)$ is a *Lie group*—a kind of differentiable manifold that has special geometry that is not preserved by generic numerical integration algorithms like 4th-order Runge-Kutta, which operate on the n-dimensional Euclidean space $\mathbb{R}^n$. Thus, a different approach is required to faithfully simulate coordinate-free rigid body dynamics. To do this, Crouch and Grossman propose a modified 4th-order Runge-Kutta scheme, which I'll call CG4.

Represented as a system of first-order ordinary differential equations (ODEs), the equations of motion for a rigid body in space are

$$\frac{\mathrm{d}}{\mathrm{d}t} \underbrace{\begin{bmatrix} T_{\mathrm{wb}}(t) \\ \mathcal{V}_{\mathrm{b}}(t) \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} T_{\mathrm{wb}}(t)\left[\mathcal{V}_{\mathrm{b}}(t)\right] \\ \mathcal{K}\left(t, T_{\mathrm{wb}}(t), \mathcal{V}_{\mathrm{b}}(t)\right) \end{bmatrix}}_{f(x)}, \tag{2}$$

where

$$\mathcal{K}\left(t, T_{\mathrm{wb}}(t), \mathcal{V}_{\mathrm{b}}(t)\right) = \mathcal{G}_{\mathrm{b}}^{-1}\left(\left[\mathrm{ad}_{\mathcal{V}_{\mathrm{b}}(t)}\right]^{\top}\mathcal{G}_{\mathrm{b}}\mathcal{V}_{\mathrm{b}}(t) + \mathcal{F}_{\mathrm{b}}(t)\right). \tag{3}$$

In these equations,

- $T_{\mathrm{wb}} \in \mathbb{SE}(3)$ is the configuration of the body's COM frame $\{\mathrm{b}\}$ relative to the inertial world frame $\{\mathrm{w}\}$.
- $[\mathcal{V}_{\mathrm{b}}] \in \mathfrak{se}(3)$ is the matrix representation of the body twist $\mathcal{V}_{\mathrm{b}} \in \mathbb{R}^6$, where $\mathcal{V}_{\mathrm{b}} = \left[\omega_{\mathrm{b}}^{\top}, v_{\mathrm{b}}^{\top}\right]^{\top}$, where $\omega_{\mathrm{b}} \in \mathbb{R}^3$ and $v_{\mathrm{b}} \in \mathbb{R}^3$ are the angular and linear velocity, respectively, represented in the COM frame $\{\mathrm{b}\}$.
- $\mathcal{G}_{\mathrm{b}} \in \mathbb{R}^{6\times 6}$ is the body's inertia tensor; $\mathcal{G}_{\mathrm{b}}$ is a diagonal matrix with $(\mathcal{I}_{xx}, \mathcal{I}_{yy}, \mathcal{I}_{zz}, \mathfrak{m}, \mathfrak{m}, \mathfrak{m})$ along the diagonal, where $\mathcal{I}_{xx}$, $\mathcal{I}_{yy}$, and $\mathcal{I}_{zz}$ are the inertias about the principal axes and $\mathfrak{m}$ is the body's mass.
- $\mathcal{F}_{\mathrm{b}} \in \mathbb{R}^6$ is the net external wrench on the body, expressed in the COM frame. $\mathcal{F}_{\mathrm{b}} = \left[m_{\mathrm{b}}^{\top}, f_{\mathrm{b}}^{\top}\right]^{\top}$, where $m_{\mathrm{b}} \in \mathbb{R}^3$ and $f_{\mathrm{b}} \in \mathbb{R}^3$ are the moment and force, respectively, in the COM frame.
- $\left[\mathrm{ad}_{\mathcal{V}_{\mathrm{b}}(t)}\right]$ is a 6x6 matrix used to express the rate of change of spatial momentum $\mathcal{P}_{\mathrm{b}} = \mathcal{G}_{\mathrm{b}}\mathcal{V}_{\mathrm{b}}$ due to the twist $\mathcal{V}_{\mathrm{b}}$.

Note that the state space of the first-order ordinary differential equation (ODE) above is $\mathcal{X} = \mathbb{SE}(3) \times \mathbb{R}^6$, and the state $x \in \mathcal{X}$ is the pair $(T_{\mathrm{wb}}, \mathcal{V}_{\mathrm{b}})$.

1.2 Discrete Dynamics

1.2.1 Event Function and Event Detection

As stated above, our event function $h : \mathbb{SE}(3) \times \mathbb{R}^6 \to \mathbb{R}$ is the distance between the bottom of the ball and the ground. If, while integrating Equation (2) from t_k to t_{k+1}, we observe that $h(x(t_{k+1})) \neq h(x(t_k))$, we must find the event time $t_{\mathrm{e}} \in [t_k, t_{k+1}]$ such that $h(x(t_{\mathrm{e}}) = 0)$.

Consider two approaches to event finding:

- Bisection root finding;
- Newton-Raphson root finding.

While the bisection method may take more iterations to converge to an acceptable estimate of t_{e}, it is guaranteed to return an estimate in the time window $[t_k, t_{k+1}]$. Conversely, while the Newton-Raphson method may be more elegant, it is possible it will return an estimate outside $[t_k, t_{k+1}]$, if $\mathrm{d}h/\mathrm{d}t$ is sufficiently small and $\mathrm{d}^2h/\mathrm{d}t^2$ is sufficiently large.

Searching for roots of the event function via bisection is simple:

1. Given a time window $[t_k, t_{k+1}]$ for which $\mathrm{sign}(h(t_k)) \neq \mathrm{sign}(h(t_{k+1}))$, divide the time window in half to produce two new time windows: $[t_k, t_k + \Delta t/2]$ and $[t_k + \Delta t/2, t_{k+1}]$, where $\Delta t = t_{k+1} - t_k$.
2. If $\mathrm{sign}(h(t_k)) \neq \mathrm{sign}(h(t_k + \Delta t/2))$, repeat step 1 for the new time window $[t_k, t_k + \Delta t/2]$. Otherwise, repeat step 1 for the new time window $[t_k + \Delta t/2, t_{k+1}]$.
3. Iterate until some convergence criterion is satisfied, or until the maximum number of iterations is reached.

Theme 1: Ellipsoid-on-Plane Impact We first describe impact detection for an ellispoid. We want to know the distance between the lowest point on an ellipsoid and a plane, as shown in Figure 1. The plane is defined by the implicit equation

$$\hat{\mathbf{n}}^{\top}\left(\mathbf{x} - \mathbf{x}_0\right), \tag{4}$$

where the point $\mathbf{x}_0$ and the unit normal $\hat{\mathbf{n}}$ are assumed to be known. For example, when $\hat{\mathbf{n}} = [0, 0, 1]^{\top}$ and $\mathbf{x}_0 = [0, 0, 0]^{\top}$, the plane is simply the x-y plane.

The ellipsoid is defined by the implicit equation

$$\left(\mathbf{x} - \mathbf{p}_{\mathrm{wb}}\right)^{\top} R_{\mathrm{wb}} D R_{\mathrm{wb}}^{\top}\left(\mathbf{x} - \mathbf{p}_{\mathrm{wb}}\right) = 1, \tag{5}$$

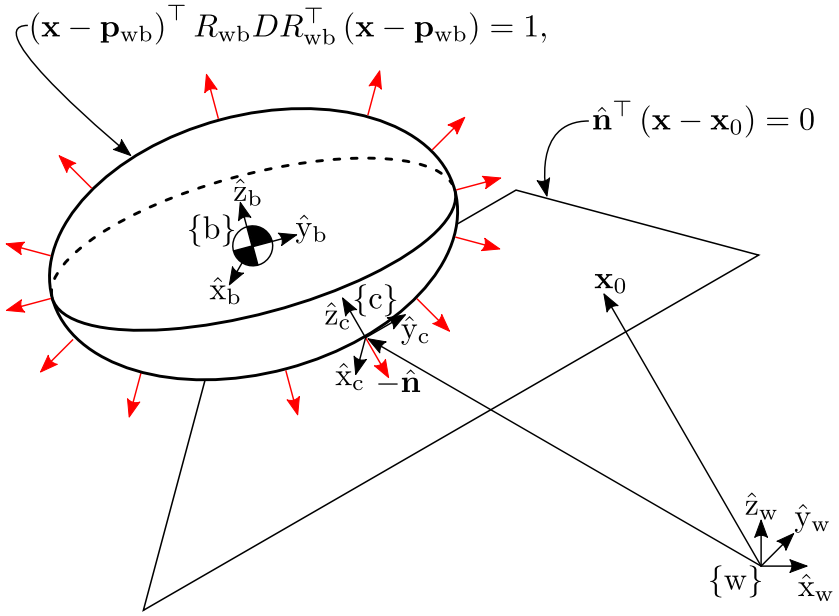


Figure 1: Ellipsoid-plane contact kinematics. The plane is defined by $\hat{\mathbf{n}}^{\top} (\mathbf{x}-\mathbf{x}_0) = 0$. The ellipsoid is defined by $(\mathbf{x}-\mathbf{p}_{\mathrm{wb}})^{\top} R_{\mathrm{wb}} D R_{\mathrm{wb}}^{\top} (\mathbf{x}-\mathbf{p}_{\mathrm{wb}}) = 1$. Red arrows extending from the ellipsoid surface indicate unit normal vectors. Contact occurs when the point $\mathbf{c}$ on the ellipse with corresponding unit normal $-\hat{\mathbf{n}}$ also satisfies $\hat{\mathbf{n}}^{\top} (\mathbf{c}-\mathbf{x}_0) = 0$. The point $\mathbf{c}$ is the origin of a frame $\{\mathrm{c}\}$. The orientation of $\{\mathrm{c}\}$ relative to $\{\mathrm{w}\}$ is any $R_{\mathrm{wc}} \in \mathbb{SO}(3)$ such that $\hat{\mathbf{z}}_{\mathrm{c}} = \hat{\mathbf{n}}$.

where $\mathbf{x} \in \mathbb{R}^3$, $(R_{\mathrm{wb}}, \mathbf{p}_{\mathrm{wb}}) \in \mathbb{SE}(3)$, and $D \in \mathbb{R}^{3\times 3}$ is a diagonal matrix whose entries are the reciprocals of the squares of the semi-axes. For convenience, we will write Equation (5) in the following more concise form:

$$\mathbf{y}^{\top} A \mathbf{y} = 1, \tag{6}$$

where $\mathbf{y} = \mathbf{x} - \mathbf{p}_{\mathrm{wb}}$ and $A = R_{\mathrm{wb}} D R_{\mathrm{wb}}^{\top}$. Note that A is symmetric. We will formulate the event function in a more general setting—how close is the ellipsoid to a flat surface whose unit normal vector is $\hat{\mathbf{e}}$?—and we will then substitute the unit normal vector corresponding to level ground, $\hat{\mathbf{e}} = -\hat{\mathbf{z}}_{\mathrm{w}} = [0, 0, -1]^{\top}$.

For an ellipsoid, the unit vector normal to the surface at the point $\mathbf{y}$ furthest in the $\hat{\mathbf{e}}$-direction is simply $\hat{\mathbf{e}}$. The matrix A in Equation (6) stretches the vector $\mathbf{y}$ by some amount μ and rotates it to align with $\hat{\mathbf{e}}$:

$$A\mathbf{y} = \mu \hat{\mathbf{e}}. \tag{7}$$

Solving for $\mathbf{y}$, we have

$$\mathbf{y} = \mu A^{-1} \hat{\mathbf{e}}. \tag{8}$$

Substituting this result into Equation (6), we have

$$\begin{aligned} 1 &= \mathbf{y}^{\top} A \mathbf{y} \\ &= \left(\mu A^{-1} \hat{\mathbf{e}}\right)^{\top} A \left(\mu A^{-1} \hat{\mathbf{e}}\right) \\ &= \mu^2 \hat{\mathbf{e}}^{\top} A^{-\top} A A^{-1} \hat{\mathbf{e}} \\ &= \mu^2 \hat{\mathbf{e}}^{\top} A^{-1} \hat{\mathbf{e}}. \end{aligned}$$

The simplification leading to the last line is possible because A is symmetric. Solving for the scaling factor μ, we get

$$\mu = \frac{1}{\sqrt{\hat{\mathbf{e}} A^{-1} \hat{\mathbf{e}}}}. \tag{9}$$

Substituting this expression for μ into Equation (8), we get

$$\mathbf{y} = \frac{A^{-1} \hat{\mathbf{e}}}{\sqrt{\hat{\mathbf{e}} A^{-1} \hat{\mathbf{e}}}}. \tag{10}$$

To find the point on the ellipsoid closest to the plane, we substitute $\hat{\mathbf{e}} = -\hat{\mathbf{n}}$ and $\mathbf{y} = \mathbf{c} - \mathbf{p}_{\mathrm{wb}}$ into Equation (10) and solve for $\mathbf{c}$:

$$\mathbf{c} = \mathbf{p}_{\mathrm{wb}} - \frac{R_{\mathrm{wb}} D^{-1} R_{\mathrm{wb}}^{\top} \hat{\mathbf{n}}}{\sqrt{\hat{\mathbf{n}}^{\top} R_{\mathrm{wb}} D^{-1} R_{\mathrm{wb}}^{\top} \hat{\mathbf{n}}}}. \tag{11}$$

The signed distance between $\mathbf{c}$ and the plane defined by Equation (4) is our event function:

$$h\left(T_{\mathrm{wb}}\right) = \hat{\mathbf{n}}^{\top}\left(\mathbf{c} - \mathbf{x}_0\right). \tag{12}$$

Contact occurs when $h(T_{\mathrm{wb}}) = 0$.

Variation 1: Superellipsoid-on-Plane Impact We introduce the superellipsoid as a smooth approximation of a rectangular prism. A superellipsoid is a generalization of an ellipsoid in the following sense:

$$\text{ellipsoid: } f(\mathbf{y}) = \sum_{j=1}^{3} \left(\frac{y_j}{r_j}\right)^2 - 1 = 0 \tag{13}$$

$$\text{superellipsoid: } f(\mathbf{y}) = \sum_{j=1}^{3} \left(\frac{x_j}{r_j}\right)^{2/\varepsilon} - 1 = 0, \tag{14}$$

where $\mathbf{y} = (y_1, y_2, y_3)$ is an arbitrary point in three-dimensional space and r_1, r_2, and r_3 are the semi-axis lengths of the ellipsoid. Increasing the smoothness parameter ε from 0 up to 1 has the effect of smoothing the corners of a rectangular prism until it becomes an ellipsoid. Increasing ε above 1 results in concave surfaces and is not considered further in this document.

Equation (13) produces a superellipsoid centered at the origin and oriented along the principal axes. The surface of a superellipsoid under a rigid-body transformation $T = (R, p)$, $T : \mathbf{y} \mapsto \mathbf{x}$ is found by inverting the rigid body transformation so as to map $\mathbf{x}$ to $\mathbf{y}$ and then applying the equation of the untransformed superellipsoid:

$$f(\mathbf{x}) = \sum_{i=1}^{3} \left(\frac{1}{r_i} \sum_{j=1}^{3} R_{ji}\left(y_j - p_j\right)\right)^{2/\varepsilon} - 1 = 0. \tag{15}$$

1.2.2 Impact Map

The impact map is derived from conservation of momentum. Equation (10) gives the contact point $\mathbf{c} \in \mathbb{R}^3$. The transformation matrix $T_{\mathrm{wc}} \in \mathbb{SE}(3)$ gives the location of the contact point in $\{\mathrm{w}\}$ and the orientation $R_{\mathrm{wc}} \in \mathbb{SO}(3)$ of $\hat{\mathbf{n}}$, the unit vector normal to the plane at $\mathbf{c}$. Note that R_{wc} is somewhat arbitrary: it can be any rotation matrix such that $\hat{z}_{\mathrm{c}} = \hat{\mathbf{n}}$:

$$R_{\mathrm{wc}} = \exp\left([\hat{\mathbf{n}}]\right),$$

where $\exp(\cdot) : \mathfrak{so}(3) \to \mathbb{SO}(3)$ is the matrix exponential and $[\hat{\mathbf{n}}] \in \mathfrak{so}(3)$ is the skew-symmetric matrix representation of $\hat{\mathbf{n}} \in \mathbb{R}^3$.

Upon impact, the rigid body experiences an impulse with magnitude λ in the direction $\hat{\mathbf{n}}$. Expressed in $\{\mathrm{c}\}$, this impulse is $\mathcal{J}_{\mathrm{c}} = \lambda \hat{\mathbf{n}}_{\mathrm{c}}$ where $\hat{\mathbf{n}}_{\mathrm{c}} = [0_{1\times 5}, 1]^{\top}$. The adjoint representation of T_{cb} is used to express this impulse in $\{\mathrm{b}\}$, where $T_{\mathrm{cb}} = T_{\mathrm{wc}}^{-1} T_{\mathrm{wb}}$:

$$\mathcal{J}_{\mathrm{b}} = \lambda \underbrace{\left[\mathrm{Ad}_{T_{\mathrm{cb}}}\right]^{\top} \hat{\mathbf{n}}_{\mathrm{c}}}_{\hat{\mathbf{n}}_{\mathrm{b}}}. \tag{16}$$

From conservation of momentum, we have

$$\mathcal{G}_{\mathrm{b}} \mathcal{V}_{\mathrm{b}}^{+} = \mathcal{G}_{\mathrm{b}} \mathcal{V}_{\mathrm{b}}^{-} + \mathcal{J}_{\mathrm{b}}, \tag{17}$$

where $\mathcal{V}_{\mathrm{b}}^{-} \in \mathbb{R}^6$ and $\mathcal{V}_{\mathrm{b}}^{+} \in \mathbb{R}^6$ are the body twist before and after impact, respectively, and where the impulse $\mathcal{J}_{\mathrm{b}}$ is given in Equation (16). We rearrange Equation (17) to express the post-impact body twist in terms of the pre-impact body twist, the as-yet-unknown impulse magnitude, and the body's spatial inertia matrix:

$$\mathcal{V}_{\mathrm{b}}^{+} = \mathcal{V}_{\mathrm{b}}^{-} + \lambda \mathcal{G}_{\mathrm{b}}^{-1} \hat{\mathbf{n}}_{\mathrm{b}}, \tag{18}$$

where $\hat{\mathbf{n}}_\mathrm{b} = [\mathrm{Ad}_{T_\mathrm{cb}}]^\top \hat{\mathbf{n}}_\mathrm{c} \in \mathbb{R}^6$. The system of equations is closed upon introducing the coefficient of restitution $\epsilon \in [0, 1]$:

$$\epsilon = -\frac{\hat{\mathbf{n}}_\mathrm{c}^\top \mathcal{V}_\mathrm{c}^+}{\hat{\mathbf{n}}_\mathrm{c}^\top \mathcal{V}_\mathrm{c}^-} = -\frac{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{V}_\mathrm{b}^+}{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{V}_\mathrm{b}^-}. \tag{19}$$

Note that $\epsilon = 0$ represents a perfectly inelastic collision and $\epsilon = 1$ represents a perfectly elastic collision. Substituting Equation (18) into Equation (19) yields the impulse magnitude:

$$\lambda = -(1+\epsilon)\frac{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{V}_\mathrm{b}^-}{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{G}_\mathrm{b}^{-1} \hat{\mathbf{n}}_\mathrm{b}}. \tag{20}$$

Substituting this solution into Equation (18) yields the post-impact body twist:

$$\mathcal{V}_\mathrm{b}^+ = \mathcal{V}_\mathrm{b}^- - (1+\epsilon)\frac{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{V}_\mathrm{b}^- \mathcal{G}_\mathrm{b}^{-1} \hat{\mathbf{n}}_\mathrm{b}}{\hat{\mathbf{n}}_\mathrm{b}^\top \mathcal{G}_\mathrm{b}^{-1} \hat{\mathbf{n}}_\mathrm{b}}. \tag{21}$$

2 Results

2.1 The Intermediate-Axis Theorem (Dzhanibekov Effect)

Before illustrating results for impact simulations, we first illustrate the intermediate-axis theorem, also known as the tennis-racket effect or the Dzhanibekov effect.

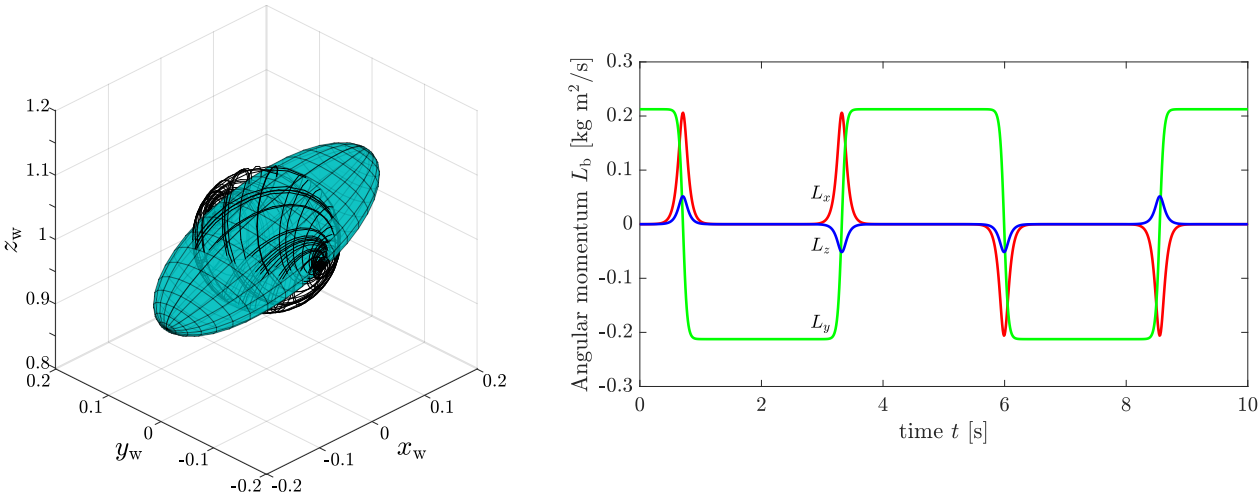


Figure 2: Illustration of the Dzhanibekov effect: in the absence of external torques, a rigid body spinning about its intermediate principal inertia axis (the y_b axis in this figure) is unstable, and it alternates between opposite orientations along this axis. Left: trace of the point on the ellipsoid furthest in the $\hat{y}_\mathrm{b}$ direction for 100 seconds of simulation. Right: angular momentum component L_y alternates between +0.25 kg-m/s and -0.25 kg-m/s. The ellipsoid has mass $\mathfrak{m}_\mathrm{b} = 1$ kg, and principal semi-axis lengths of 0.05 m, 0.1, and 0.2 m along $\hat{x}_\mathrm{b}$, $\hat{y}_\mathrm{b}$, and $\hat{z}_\mathrm{b}$, respectively. Simulator timestep is 0.01 seconds, and $\mathcal{V}_{\mathrm{b}(0)} = [0.001, 25, 0.001, 0, 0, 0]^\top$ (SI units).

2.2 A Bouncing Football

We present results for a bouncing ellipsoid that resembles an American football.

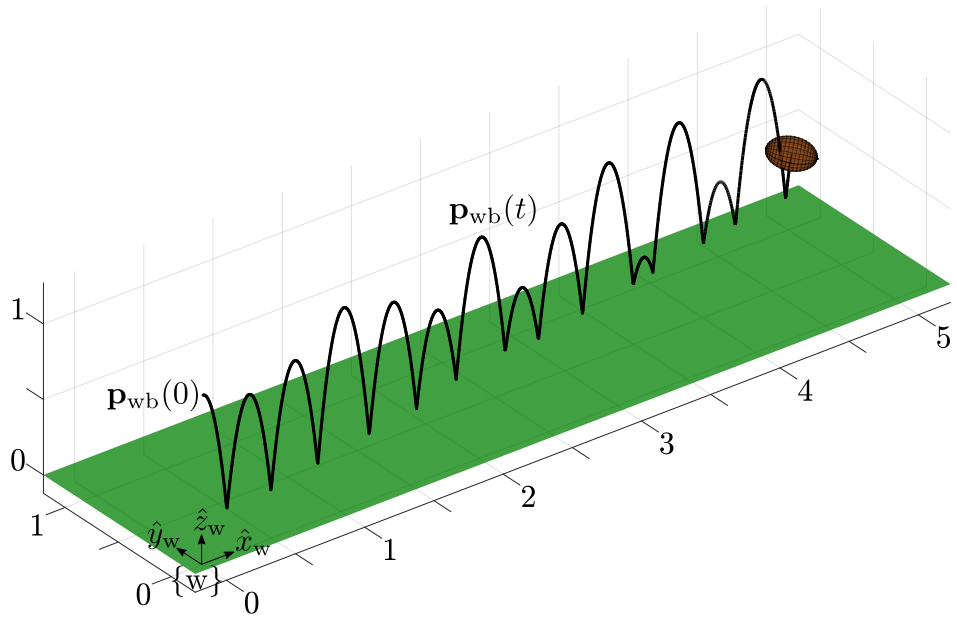


Figure 3: COM trajectory $\mathbf{p}_{\mathrm{wb}}(t)$ for a bouncing football simulated with a timestep of 0.01 seconds and perfectly elastic collisions, i.e., $\epsilon = 1$. Initial conditions: $R_{\mathrm{wb}}(0) = I_{3\times 3}$, $\mathbf{p}_{\mathrm{wb}}(0) = [0, 0, 1]^\top$, $\mathcal{V}_{\mathrm{b}}(0) = [10, 10, 10, 0.5, 0.1, 0]^\top$ (SI units). The football has mass $\mathfrak{m}_{\mathrm{b}} = 1$ kg and principal semi-axis lengths of 0.1 m, 0.1, and 0.15 m along $\hat{x}_{\mathrm{b}}$, $\hat{y}_{\mathrm{b}}$, and $\hat{z}_{\mathrm{b}}$, respectively. Acceleration due to gravity is 9.81 m/s^2.